

Prototype 1: Underwater Glider Design

Philip Esclamado
Lead Engineer
University of Victoria Environmental Engineering Club

May 26, 2023

1 Introduction

This report presents the design and development of a pool operation system, focusing on buoyancy manipulation and vehicle dynamics. The system aims to provide precise control and maneuverability within the pool environment, catering to the specific needs of pool operators. By adhering to established design criteria and requirements, the objective is to create an Autonomous Underwater Vehicle (AUV) that meets the identified needs and expectations.

The report outlines the key components and design considerations of the pool operation system, starting with an overview of the design criteria and requirements. These criteria serve as benchmarks for evaluating the system's performance and guide the design process to ensure the system's effectiveness and efficiency.

2 Design Criteria and Requirements

The system should incorporate effective buoyancy manipulation capabilities, enabling precise control for smooth maneuverability and stability of the vehicle within the pool environment. Vehicle dynamics should be prioritized to ensure efficient movement and navigation, allowing the vehicle to achieve desired speeds, accelerations, and turning radii while maintaining stability and control.

Compatibility with standard pool dimensions and conditions is essential, ensuring safe and effective operation in pools of varying depths, temperature ranges, and water conditions. The design must prioritize safety, incorporating features such as collision avoidance mechanisms, emergency stop controls, and fail-safe systems to protect pool users and the surrounding environment. An intuitive and user-friendly interface should be provided, allowing operators to monitor and adjust vehicle parameters, including speed, direction, and buoyancy, in real-time.

Durability and maintenance considerations should be integrated into the design, ensuring the vehicle can withstand pool conditions and regular usage. Easy accessibility to critical components for inspection, repair, or replacement should be facilitated. Power utilization and energy efficiency should be optimized, providing sufficient battery life or power source to support extended periods of operation while minimizing recharge or replacement requirements.

Cost-effectiveness should be a key consideration, utilizing readily available and affordable materials, components, and technologies. The design should also allow for scalability and adaptability to accommodate potential future enhancements or modifications. Regulatory compliance with safety, environmental, and relevant pool operation standards and guidelines must be ensured.

3 Buoyancy Engine

Two check valves are strategically placed within the system to regulate the flow of the working fluid. The first check valve is positioned at the intake of the bladder, allowing the fluid to enter the bladder when the pressure inside is lower than the external pressure. This valve ensures that the bladder is filled with the working fluid during the initial stage of operation.

The second check valve is placed at the outlet of the bladder, preventing the fluid from flowing back into the surrounding environment once it enters the bladder. This valve ensures that the working fluid remains contained within the bladder and maintains the desired buoyancy.

Additionally, a valve is included in the design to control the release of the working fluid from the bladder. This valve enables precise control over the buoyancy force generated by the engine, allowing for adjustments to suit specific requirements. By manipulating the valve, the operator can regulate the amount of fluid released from the bladder, consequently altering the buoyancy and the resulting force.

To facilitate the operation of the system, a pump is integrated into the design. The pump is responsible for pressurizing the working fluid and supplying it to the bladder. By activating the pump, the operator can ensure a continuous flow of fluid into the bladder, maintaining the buoyancy and power output of the engine.

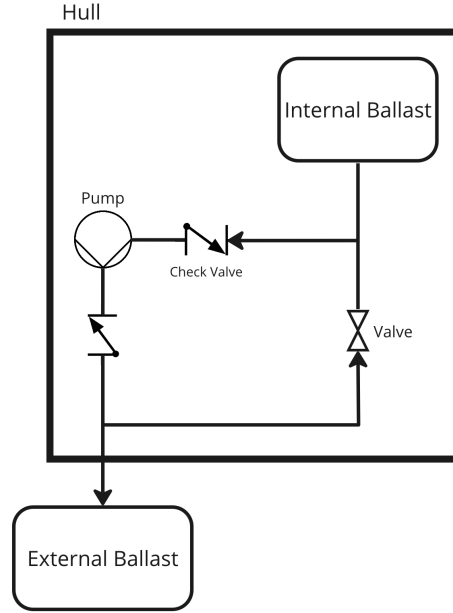


Figure 1: Bladder design piping diagram for glider propulsion [1].

4 Housing

The main body of the housing will be constructed using an acrylic tube. Acrylic is selected due to its favourable chemical compatibility with chlorinated water within the typical range of 5-10 ppm [2]. This ensures that the housing remains resistant to chemical degradation and maintains its structural integrity throughout the system's operation within the pool.

To secure the ends of the acrylic tube and provide structural support, Delrin endcaps will be employed. Delrin is a durable engineering plastic known for its mechanical strength and stability. However, it is important to note that Delrin is not recommended for any use involving contact with chlorinated water within 5-10 ppm, as it can experience severe adverse effects [3].

5 Raspberry Pi 4

The Raspberry Pi 4 will serve as the central control unit, responsible for executing control algorithms, interfacing with sensors and actuators, and managing the overall operation of the pool system. Its GPIO (General-Purpose Input/Output) pins will be utilized to connect and control peripheral devices, such as sensors, motors, and valves, enabling precise and coordinated control actions.

6 Control System

The closed-loop control system operates by continuously measuring the actual depth of the vehicle using depth sensors and comparing it to the target depth set by the operator. Any deviation between the actual and desired depths is then used as feedback to adjust the control inputs.

To achieve this, a proportional-integral-derivative (PID) controller will be employed. The PID controller calculates and applies control actions based on the error signal obtained from the depth measurements. The proportional term provides immediate response to the error, the integral term eliminates steady-state errors, and the derivative term accounts for the rate of change of the error, improving the system's stability and responsiveness.

The control system will receive depth measurements from the sensors and generate control signals to actuate the buoyancy manipulation mechanism.

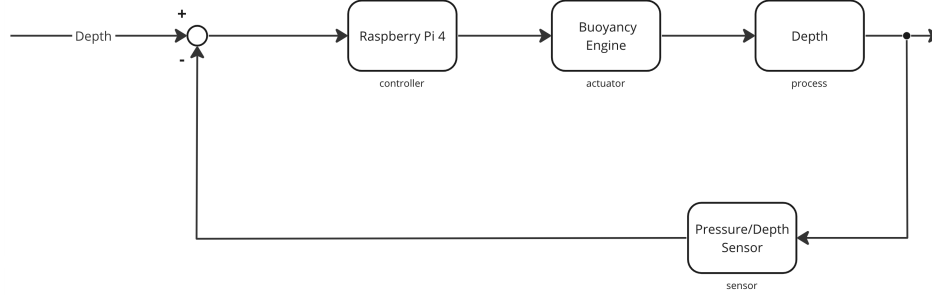


Figure 2: Closed-loop control scheme for depth control.

7 Power

The Raspberry Pi requires a power input of 5 V and 3 A, which will be supplied through GPIO pins. Furthermore, a system budget of 20 V has been allocated to accommodate the entire setup, considering that the pumps typically used in this operation operate at a maximum voltage of 12-15 V. To meet the power needs, the design will utilize rechargeable batteries, ensuring a sustainable and reliable power source for the system's operation.

7.1 Buck Voltage Converter

In this design, the buck converter will be connected to the power source supplying the system. It will convert the input voltage, which is typically 20 V, to the desired lower voltage of 5 V for the Raspberry Pi 4. Simultaneously, the buck converter will regulate the voltage supplied to the pump, ensuring that it remains within the specified operating range, typically between 12 V and 15 V.

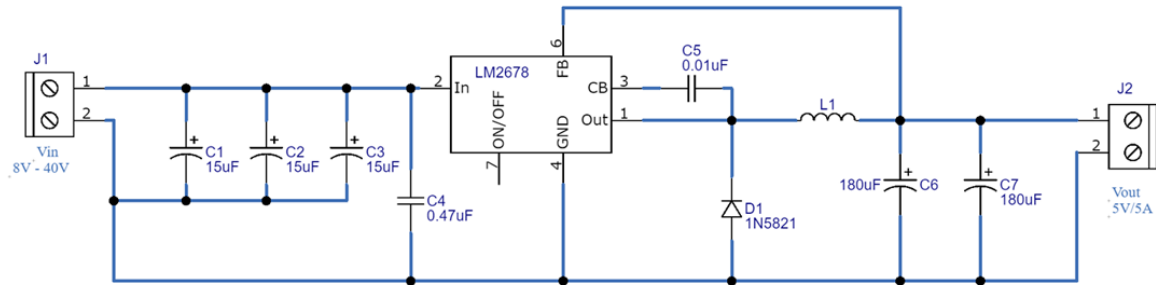


Figure 3: Step-Down Voltage Regulator Circuit.

8 Recovery

To ensure the safe and effective recovery of the AUV, a diver-activated kill switch has been implemented, following a similar mounting method as the BlueRobotics WetLink penetrators seen in Figure 4. The recovery mechanism now involves securely mounting a BlueRobotics underwater switch on the end cap. This switch can be manually triggered by a diver to initiate the desired action. By incorporating this underwater switch, the AUV's retrieval process becomes more reliable and efficient.

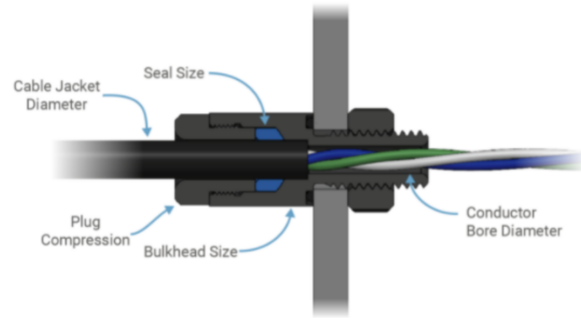


Figure 4: BlueRobotics WetLink Penetrator Diagram [4].

References

- [1] “Buoyancy engine,” https://en.wikipedia.org/wiki/Buoyancy_engine, accessed: May 18, 2023.
- [2] Industrial Specialties Mfg., “Acrylic (PMMA) Chemical Compatibility Chart,” <https://marketing.industrialspec.com/acton/attachment/30397/f-006b/1/-/-/-/-/ acrylic-pmma-chemical-compatibility-chart-from-ism.pdf>, n.d.
- [3] Industrial Specialties Manufacturing, “Acetal (POM) Chemical Compatibility Chart,” <https://www.industrialspec.com/images/files/acetal-pom-chemical-compatibility-chart-from-ism.pdf>, accessed: May 18, 2023.
- [4] “Blue robotics wlp-vp,” <https://bluerobotics.com/store/cables-connectors/penetrators/wlp-vp/>, accessed: May 26, 2023.
- [5] skhg, “raspberrypi-killswitch,” <https://github.com/skhg/raspberrypi-killswitch>.
- [6] “When Should You Choose LDO or Buck Converter?” <https://www.circuitbread.com/ee-faq/when-should-you-choose-ldo-or-buck-converter>, accessed: May 18, 2023.
- [7] “Switch 10.5A R1,” <https://bluerobotics.com/store/comm-control-power/switch/switch-10-5a-r1/>, accessed: May 18, 2023.